

Database System Engineering and Distributed Backend Development Course

Code: 25CS1302E

ROLL NO : 2520030255

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WEEK-3 SQL JOIN TRANSCATIONS

```
mysql> CREATE DATABASE join2;
Query OK, 1 row affected (0.01 sec)

mysql> USE join2;
Database changed
mysql> CREATE TABLE roles(
    ->     role_id INT PRIMARY KEY,
    ->     role_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE users(
    ->     user_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     email VARCHAR(50),
    ->     role_id INT
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE departments(
    ->     dept_id INT PRIMARY KEY,
    ->     dept_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE employees(
    ->     emp_id INT PRIMARY KEY,
    ->     emp_name VARCHAR(50),
    ->     dept_id INT,
    ->     manager_id INT,
    ->     salary DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE customers(
    ->     customer_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(50),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE orders(
    ->     order_id INT PRIMARY KEY,
    ->     customer_id INT,
    ->     order_date DATE,
    ->     total_amount DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```

mysql> CREATE TABLE accounts(
    ->     account_no INT PRIMARY KEY,
    ->     holder_name VARCHAR(50),
    ->     balance DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO roles VALUES
    -> (1,'Admin'),
    -> (2,'Manager'),
    -> (3,'Employee'),
    -> (4,'HR');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO users VALUES
    -> (101,'Rahul','rahul@gmail.com',1),
    -> (102,'Anjali','anjali@gmail.com',2),
    -> (103,'Ravi','ravi@gmail.com',3),
    -> (104,'Sneha','sneha@gmail.com',NULL);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO departments VALUES
    -> (10,'CSE'),
    -> (20,'ECE'),
    -> (30,'EEE'),
    -> (40,'MECH');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO employees VALUES
    -> (1,'Amit',10,NULL,70000),
    -> (2,'Rohit',10,1,55000),
    -> (3,'Kiran',20,1,45000),
    -> (4,'Priya',30,2,60000),
    -> (5,'Neha',10,2,52000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO customers VALUES
    -> (1,'Raj','Hyderabad'),
    -> (2,'Arjun','Chennai'),
    -> (3,'Meena','Bangalore');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

```

```
mysql> INSERT INTO customers VALUES
    -> (1, 'Raj', 'Hyderabad'),
    -> (2, 'Arjun', 'Chennai'),
    -> (3, 'Meena', 'Bangalore');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO orders VALUES
    -> (1001, 1, '2026-08-01', 1500),
    -> (1002, 2, '2026-08-02', 2500),
    -> (1003, 1, '2026-08-03', 3000),
    -> (1004, 3, '2026-08-04', 1800);
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO accounts VALUES
    -> (101, 'Rahul', 50000),
    -> (102, 'Priya', 40000),
    -> (103, 'Kiran', 35000);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT emp_name, dept_name
    -> FROM employees e
    -> INNER JOIN departments d
    -> ON e.dept_id = d.dept_id;
```

emp_name	dept_name
Amit	CSE
Rohit	CSE
Kiran	ECE
Priya	EEE
Neha	CSE

5 rows in set (0.00 sec)

```
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	email	role_name
Rahul	rahul@gmail.com	Admin
Anjali	anjali@gmail.com	Manager
Ravi	ravi@gmail.com	Employee
Sneha	sneha@gmail.com	NULL

4 rows in set (0.00 sec)

```
mysql> SELECT name, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	role_name
Rahul	Admin
Anjali	Manager
Ravi	Employee
NULL	HR

4 rows in set (0.00 sec)

```
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id
->
-> UNION
->
-> SELECT name, email, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	email	role_name
Rahul	rahul@gmail.com	Admin
Anjali	anjali@gmail.com	Manager
Ravi	ravi@gmail.com	Employee
Sneha	sneha@gmail.com	NULL
NULL	NULL	HR

```
mysql> SELECT e.emp_name AS Employee,
->          m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id = m.emp_id;
```

Employee	Manager
Amit	NULL
Rohit	Amit
Kiran	Amit
Priya	Rohit
Neha	Rohit

5 rows in set (0.00 sec)

```
mysql> SELECT o.order_id,
->          customer_name,
->          city,
->          total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id;
```

order_id	customer_name	city	total_amount
1001	Raj	Hyderabad	1500.00
1002	Arjun	Chennai	2500.00
1003	Raj	Hyderabad	3000.00
1004	Meena	Bangalore	1800.00

4 rows in set (0.00 sec)

```
mysql> SELECT dept_name,
->          COUNT(*) AS Employee_Count,
->          AVG(salary) AS Average_Salary,
->          MAX(salary) AS Maximum_Salary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY dept_name;
```

dept_name	Employee_Count	Average_Salary	Maximum_Salary
CSE	3	59000.000000	70000.00
ECE	1	45000.000000	45000.00
EEE	1	60000.000000	60000.00

```
mysql> SELECT emp_name,
->          salary,
->          dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> WHERE salary > 50000
-> AND dept_name = 'CSE';
```

emp_name	salary	dept_name
Amit	70000.00	CSE
Rohit	55000.00	CSE
Neha	52000.00	CSE

3 rows in set (0.00 sec)

```
mysql> SELECT customer_name,
->          SUM(total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id
-> GROUP BY customer_name
-> ORDER BY TotalPurchase DESC;
```

customer_name	TotalPurchase
Raj	4500.00
Arjun	2500.00
Meena	1800.00

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE faculty(
->     faculty_id INT PRIMARY KEY,
->     faculty_name VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
mysql> CREATE TABLE courses(
->     course_id INT PRIMARY KEY,
->     course_name VARCHAR(50),
->     faculty_id INT
-> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> INSERT INTO faculty VALUES
-> (1,'Dr. Rao'),
-> (2,'Dr. Kumar');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO courses VALUES
-> (101,'DBMS',1),
-> (102,'Java',2);
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO students VALUES
-> (1,'Rahul'),
-> (2,'Sneha');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO enrollments VALUES
-> (1,1,101),
-> (2,2,102);
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT student_name,
->           course_name,
->           faculty_name
-> FROM students s
-> JOIN enrollments e
-> ON s.student_id = e.student_id
-> JOIN courses c
-> ON e.course_id = c.course_id
-> JOIN faculty f
-> ON c.faculty_id = f.faculty_id;
```

```
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Rahul        | DBMS        | Dr. Rao      |
| Sneha        | Java        | Dr. Kumar    |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

